C12.3 DVM transaction process and rules

This section describes the sequencing and other rules for each DVM transaction.

C12.3.1 DVM Operation process

1. The originating master component issues the DVM Operation transaction on its read address channel.
2. The interconnect component distributes the transaction to all participating components using the appropriate snoop address channel.
3. Each participating component acknowledges receipt of the message using the snoop response channel.
4. The interconnect component collects the acknowledgements and responds to the original DVM transaction using the read data channel of the originating master component.

C12.3.2 DVM Sync and DVM Complete transactions

1. The originating master component issues the DVM Sync on its read address channel.
2. The interconnect component distributes the DVM Sync to all participating components using the appropriate snoop address channel.
3. Each participating component acknowledges receipt of the DVM Sync using the snoop response channel.
4. The interconnect component collects the acknowledgements and responds to the original DVM Sync using the read data channel of the originating master component.
5. Each participating component must issue a DVM Complete when it has completed all the necessary actions. The DVM Complete is issued by each participating component, using its read address channel. A DVM Complete on the read address channel must only be issued after the handshake of the associated DVM Sync on the snoop address channel of the same master.
6. The interconnect component can respond immediately to the DVM Complete transaction using the read data channel of the component that issued the DVM Complete.
7. The interconnect component observes all of the DVM Complete transactions and when it has received a DVM Complete from each participating component, it issues a DVM Complete, using the snoop address channel of the master component that originally issued the DVM Sync.
8. The originating master component acknowledges the receipt of the DVM Complete using the snoop response channel.

C12.3.3 Multi-part DVM Operation transactions

Multi-part DVM messages are always sent as successive transactions and no other transaction can be interposed between them. A master component issuing a multi-part DVM message must be able to issue the latter parts of the message without requiring any other external actions to occur.

Each transaction of a multi-part DVM Message has a response, both on the snoop response and read data channels.

Each transaction of a multi-part DVM Message must use the same AXI ID. See *AXI transaction identifiers* on page A5-76.

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C12.3.4 Transaction response

When a component receives a DVM transaction, it must respond as follows:

- If the component can perform the requested action, it must respond by setting **CRRESP** to `0b00000`.
- If the component is unable to perform the requested action, it must respond by setting **CRRESP** to `0b00010`. Typically, this response indicates an unsupported message.

A component is not permitted to set **CRRESP** to `0b00010` in response to a DVM Sync or a DVM Complete.

The interconnect component gathers all response values and responds to the originator as follows:

- if **CRRESP** is `0b00000` for all responses, the interconnect component sets **RRESP** to `0b0000`
- If **CRRESP** is `0b00010` for any responses, the interconnect component sets **RRESP** to `0b0010`.

Note

If **CRRESP** is `0b00010` for any responses, this specification recommends that the interconnect component provides a fault log to record which components are unable to perform requested actions.

No data transfer is associated with DVM transactions.

For multi-part DVM transactions, a response is provided for each transaction. For such transactions, it is required that a component must provide the same response to each transaction.

C12.3.5 Message ID

In general, DVM transactions must not use AXI ID values that are used by non-DVM transactions on the AR channel. This ensures that there are no ordering constraints between DVM transactions and non-DVM transactions. However:

- DVM messages and non-DVM transactions can use the same AXI ID value, provided one transaction has fully completed before the other is issued.
- DVM messages are permitted to use the same AXI ID value as a transaction issued on the AW channel.
- Different DVM transactions can use either the same AXI IDs, or different ones. Each transaction of a multi-part DVM Operation must use the same AXI ID.
- If different AXI IDs are used for DVM Operations then the order of arrival of the different messages at the recipient of the message is not guaranteed. All DVM Operations must be correctly ordered with respect to a DVM Sync from the same issuing master component, even if different AXI ID values are used.

C12.3.6 Instruction cache invalidation alternatives

In general, instruction caches can use either a physical address or a virtual address to tag the data they contain. A system might contain a mixture of components meaning that both address types are used.

The DVM protocol includes instruction cache invalidation operations that use physical addresses and operations that use virtual addresses. A component is only required to broadcast one format of instruction cache invalidation, either virtual address based, or physical address based. However, a component is permitted to broadcast both types, and must correctly receive both types of invalidation. All recipients of the message must perform the appropriate action.

If the format of the message does not match the style of instruction cache implemented over-invalidation, that is, invalidation of more entries in the instruction cache than is functionally required, will be necessary to ensure that the required action is performed.

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